

MATH 2630-090: TEST 3 – Summer 2024

July 10, 2024

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Name (*please print*): _____

Instructions: Please read the following instructions carefully.

- **Read all of the directions** for each problem.
- You have **75 minutes** to finish the exam.
- **Show work to receive credit.** Answers that are wrong, but still have work might receive partial credit. If you provide the answer to a computation without work, I will not give partial credit.
- Provide **exact** solutions.
- Do **not** write in the table below.

Problem	Points	Score
1	12	
2	10	
3	15	
4	12	
5	12	
6	12	
7	15	
8	12	
Total	100	

1. (12 points) Evaluate the following double integral where R is the region bounded by the first quadrant, $x = 3$ and $y = \sqrt{x}$.

$$\iint_R \frac{y}{x^2 + 1} \, dA$$

2. (10 points) Estimate the volume under the surface $f(x, y)$, over the region $R = \{(x, y) : 0 \leq x \leq 2, 0 \leq y \leq 4\}$ using upper right corners, $m = 2$ and $n = 2$.

$$f(x, y) = \frac{xy}{x + y + 1}$$

3. (15 points) Find the volume of T , where T is the solid bounded by the xy -plane, $z = 3 + x$, $x^2 + y^2 = 4$, and $x^2 + y^2 = 1$.

4. **(12 point)** Use spherical coordinates to find an expression for the volume of the region bounded by the first octant, $x^2 + y^2 + z^2 = 2z$ and $x^2 + y^2 + z^2 = 4$. **DO NOT EVALUATE!**

5. (12 points) Consider the following double integral.

$$\iint_R f(x, y) \, dA = \int_0^1 \int_y^{\sqrt{y}} x - y \, dx \, dy$$

- (a) Reverse the order of integration.
- (b) Evaluate the integral.
- (c) If the area of the R is $\frac{1}{6}$ find the average value of $f(x, y)$.

6. (12 points) Prove using integration that the volume of the region between spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$ is $\frac{76\pi}{3}$.

7. (15 points) Set up **BUT DO NOT EVALUATE** the following double integral using the transformation $x = \frac{1}{2}(u + v)$ and $y = \frac{1}{2}(u - v)$.

$$\iint_R \frac{4}{(x - y)^2} \, dx \, dy$$

Where R is the trapezoid bounded by the lines $x - y = 2$, $x - y = 4$, $x = 0$, and $y = 0$.

8. (12 points) If $\mu(x, y) = e^{x^2+y^2}$ is the mass density function of the disc $D = \{(x, y) : x^2 + y^2 \leq 81\}$, find the mass of D .

JUST FOR FUN. (OPTIONAL)

As I mentioned in class I am creating a 3D-solid to represent each class I teach. If you want to be apart of it...

give me two numbers (x, y) where $0 \leq x \leq 3.14159$ and $0 \leq y \leq 6.28318$

Try to use 2-5 decimals so for example $(.4582, 2.9893)$ or $(1.1124, 5.0035)$